

RESUME

DATA ENGINEER | DATA SCIENTIST

FARHAN HAMID LUBIS

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PROFESSIONAL SUMMARY

Results-driven Data Engineer & MLOps Engineer combining 3+ years of data-intensive experience in oil & gas research with hands-on production engineering skills. Backed by 10+ years of Python programming and a strong quantitative foundation from scientific computing, with proven expertise in designing ETL/ELT pipelines (Airflow, Kafka, Spark), orchestrating containerized infrastructure (PostgreSQL, MLflow, Redis, Superset), and operationalizing ML models with experiment tracking, feature stores, and automated retraining. Passionate about turning complex data into scalable, reliable systems across tech and ML-heavy environments.

SKILLS

DATA ENGINEERING

Apache Airflow · Kafka · Spark · PostgreSQL

ML & MLOPS

MLflow · Scikit-learn · PyTorch · TensorFlow

INFRASTRUCTURE

Docker · Linux · AWS · FastAPI · Git · Proxmox

LANGUAGES

Python · SQL · JavaScript

EDUCATION

2026

Hacktiv8 Bootcamp

Data Science & AI Machine Learning — Score 94.32%

Jakarta, Indonesia (Online)

2020 – 2022

Bandung Institute of Technology (ITB)

M.Eng. Geophysical Engineering — GPA 3.64/4.00

Bandung, Indonesia

2014 – 2019

Universitas Padjadjaran

B.Sc. Geophysics — GPA 3.54/4.00

Bandung, Indonesia

EXPERIENCE

JUN 2024 - DEC 2025

PT. LAPI ITB — PERTAMINA TD-I

Research Assistant · Data & Simulation Engineering

- Designed end-to-end data pipelines for large-scale pore-scale simulations on multi-GB 3D rock image datasets, managing the full data lifecycle from raw CT scans to analytical outputs
- Engineered Python-based automated workflows for elastic property analysis, replacing manual processing with reproducible parameterized pipelines
- Applied data-driven numerical modeling (Python + OpenFOAM + GeoDict) for CCS/CCUS feasibility studies and reservoir characterization in collaboration with Pertamina
- Delivered cross-functional insights to academic and industry stakeholders, translating simulation outputs into actionable engineering recommendations

DEC 2023 - DEC 2024

PT. LAPI ITB — PERTAMINA EP

Geophysicist · Data Integration & Interpretation

- Integrated multi-source subsurface datasets (seismic, well log, formation evaluation) into unified interpretation workflows for West Java O&G field
- Developed data-driven interpretation pipelines to improve reservoir characterization accuracy for Low-Quality Low-Contrast reservoir targets
- Managed and processed large heterogeneous geoscience datasets using HampsonRussell, Petrel, and custom Python automation scripts

- Built Python and MATLAB automation pipelines to extract physical properties (porosity, permeability, elastic moduli) from 3D digital rock models — eliminating repetitive manual analysis steps
- Processed and managed large 3D micro-CT image datasets using scientific computing tools (GeoDict, ParaView) with HPC/MPI parallel computing infrastructure
- Presented workflow automation research at ITB & Pertamina Digital Rock Physics Summer School (Aug 2023)

Portfolio
farhanhl-ds.github.io

FEATURED PROJECTS

STAYWISE — Enterprise MLOps Platform
E-Commerce Churn Prediction & Retention Analytics

- Architected production-grade MLOps stack: PostgreSQL DW, Airflow, MLflow, Kafka, Redis; fully containerized with Docker Compose (VPS + local)
- Built Airflow DAG pipeline for RFM feature engineering and churn labeling from raw e-commerce transactions into PostgreSQL data warehouse (9+ tables, star schema)
- Integrated MLflow for experiment tracking and model versioning
- Built Next.js + FastAPI dashboard with choropleth maps for geospatial analysis
- Developed an AI-powered analytics feature leveraging streaming SSE with Ollama, transforming structured LLM-generated JSON outputs into interactive visualizations.

Tech Stack

FastAPI · Next.js · Redis · Airflow
· MLflow · PostgreSQL · Docker ·
LLM · Olama · Kafka · AWS S3

E-COMMERCE BATCH ETL PIPELINE
Amazon India Sales — 128K+ Transactions

- Developed an end-to-end batch ETL pipeline using Apache Airflow to orchestrate data ingestion from PostgreSQL, integrated with Great Expectations for data validation and Elasticsearch for downstream analytics, fully containerized with Docker
- Implemented data quality gates to catch schema drift and null violations before loading
- Designed Kibana dashboard featuring key business KPIs for sales channel analysis, regional demand monitoring, and product performance

Tech Stack

Airflow · PostgreSQL · Docker ·
Great Expectations · Elasticsearch
· Kibana

EARTHQUAKE BUILDING DAMAGE PREDICTION
Deployed ML Service — 762K Records

- Engineered multiclass classification pipeline (Random Forest) on 762K building records from 2015 Nepal earthquake; handled class imbalance and prevented data leakage
- Deployed as production-ready Streamlit app on HuggingFace Spaces supporting single and batch inference via CSV upload

Tech Stack

Scikit-learn · HuggingFace Spaces
· imbalanced-learn · Streamlit

CERTIFICATIONS

- Hacktiv8 Full-Time Data Science & AI Machine Learning Program
- HackerRank: Python (Basic), SQL (Intermediate)
- BCG: Data Science Job Simulation
- freeCodeCamp: Responsive Web Design

LANGUAGES

Indonesian - Native
English - Professional Proficiency
IELTS overall score 7.0 · B2 CEFR

REFERENCES

Wendy
Instructor · Hacktiv8
wendy@hacktiv8.com

Ali Aufa Fakhri
Instructor · Hacktiv8
aliaufafakhri@gmail.com

“I hereby declare that all information outlined above is true and correct to the best of my knowledge.”

Last updated: May 2026